



1.6

Polynomial Functions and End Behavior

Student Notes and Guided Practice

AP Precalculus - Unit 1

| ECHS Mathematics

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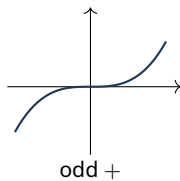
Ghuwaleh

Learning Targets

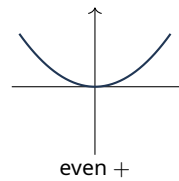
- Determine polynomial end behavior from degree and leading coefficient.
- Express end behavior in words, arrows, and limit notation.
- Distinguish global end behavior from local turning behavior.

Essential Question

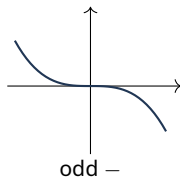
How do degree parity and leading coefficient determine the far-left and far-right behavior?



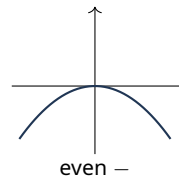
odd +



even +



odd -



even -

Concept Framework

Key Idea

Even degree means both ends point in the same vertical direction.

Key Idea

Odd degree means the ends point in opposite vertical directions.

Key Idea

The sign of the leading coefficient controls the right end.

Key Idea

End behavior does not determine the number or locations of turning points.

Formula Map

Formula and Structure

$$P(x) = a_n x^n + \cdots$$

Formula and Structure

$$P(x) \sim a_n x^n \quad \text{as } |x| \rightarrow \infty$$

Worked Examples

Worked Example 1

Problem. Describe the end behavior of $P(x) = -4x^7 + 2x^3 - 1$.

Solution. Odd degree and negative leading coefficient: $x \rightarrow -\infty \Rightarrow P(x) \rightarrow \infty$; $x \rightarrow \infty \Rightarrow P(x) \rightarrow -\infty$.

Worked Example 2

Problem. Can a positive-leading even-degree polynomial have a local maximum?

Solution. Yes. Both ends rise, but the graph may turn multiple times between the ends.

Worked Example 3

Problem. Compare $3x^6 - 10x^5$ and $3x^6 + 1000x$ for large $|x|$.

Solution. Both are asymptotic in relative behavior to $3x^6$, so both ends rise.

Multiple-Representation Connection

AP Reasoning

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Describe both ends of $4x^8 - 3x + 1$ in words and limit notation.

2. Describe both ends of $-2x^5 + x^4$ in words and limit notation.

3. Describe both ends of $-x^{10} + 100x^9$ in words and limit notation.

4. Describe both ends of $7 - 3x^3$ in words and limit notation.

5. Describe both ends of $0.2x^6 - 10x^2$ in words and limit notation.

6. Two polynomials have the same degree parity and leading coefficient sign. What end behavior must they share, and what may differ?

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.

Video support: [Flipped Math topic page](#) | Original ECHS material informed by Pearson and Holt Precalculus concepts.